

Initial Analysis Document

HealthConnect Clinic — Appointment Attendance & No-Show Analysis

Track	Data Analytics
Project	HealthConnect Clinic Experience Lab
Deliverable	Week 4 Initial Analysis Document
Data Sources	HealthConnect_Appointment_Data.csv; HealthConnect_Data_Dictionary.xlsx
Records Reviewed	5,000 appointment records across 18 variables

1. Problem Understanding and Track Role

HealthConnect Clinic is experiencing a high rate of missed appointments (no-shows), which reduces the efficient use of appointment slots and limits the clinic's ability to serve patients who need care. The clinic wants to understand why patients miss appointments and how data and AI can help reduce this problem and improve the overall patient support experience.

Within this shared project, the Data Analytics track is responsible for exploring the appointment dataset to understand patterns and relationships associated with attendance, cancellations, and no-shows. The output of this track will provide the descriptive, business-facing insight that complements the predictive work of the Data Science track and informs the Generative AI assistant's understanding of appointment-related patient behaviour.

Track-specific problem statement:

“What patterns exist in the HealthConnect appointment data that explain why patients miss appointments, and which measurable indicators (KPIs) would help the clinic monitor and reduce no-shows going forward?”

2. Resources Reviewed

- **HealthConnect_Appointment_Data.csv** — 5,000 fictional, anonymised appointment records, one row per appointment, spanning appointment dates from January 2025 to June 2026.
- **HealthConnect_Data_Dictionary.xlsx** — defines all 18 variables, their data types, and usage notes; used throughout this review to confirm variable meaning and expected value ranges before drawing conclusions.
- **HealthConnect_Clinic_Knowledge_Base.docx** — reviewed for context on clinic operations (locations, opening hours, appointment and cancellation procedures). This surfaced one discrepancy noted in Section 4 (Data Quality Assessment) between documented opening hours and the appointment dataset.

3. Dataset Overview

The dataset contains 5,000 appointment-level records with no duplicate rows and no duplicate `appointment_id` values, confirming `appointment_id` is a reliable unique identifier. Patients can appear more than once (repeat visits), so analysis at the patient level would require aggregation rather than treating each row as an independent patient.

3.1 Variable Groups

Identifiers	<code>appointment_id</code> , <code>patient_id</code>	Uniquely identify each appointment and the patient it belongs to.
Demographics	<code>gender</code> , <code>age</code> , <code>age_group</code>	Describe who the patient is.
Appointment details	<code>appointment_type</code> , <code>appointment_date</code> , <code>appointment_day</code> , <code>appointment_time</code>	Describe what was booked and when.
Booking behaviour	<code>booking_date</code> , <code>booking_lead_days</code>	Describe how far in advance the appointment was made.
Patient history	<code>previous_appointments</code> , <code>previous_no_shows</code>	Describe the patient's past attendance behaviour.
Reminders	<code>reminder_sent</code> , <code>reminder_channel</code>	Describe whether/how the patient was reminded.
Logistics	<code>distance_to_clinic_km</code> , <code>waiting_time_minutes</code>	Describe access and in-clinic experience factors.
Outcome	<code>appointment_outcome</code>	Attended, No-Show, or Cancelled — the outcome of interest.

3.2 Outcome Distribution

The `appointment_outcome` variable is the central variable for this project. Its distribution across the 5,000 records is as follows:

No-Show	2,423	48.5%
Attended	2,314	46.3%
Cancelled	263	5.3%

Nearly one in two appointments in the dataset ends in a no-show. This confirms the business scenario is well represented in the data and gives a clear baseline against which future KPIs and model performance can be compared.

4. Data Quality Assessment

4.1 Completeness (Missing Values)

reminder_channel	1,366	27.3%	Not a defect — missing exactly where reminder_sent = “No”. Should be treated as “Not Applicable” rather than a genuine gap.
distance_to_clinic_km	90	1.8%	Genuine missing values; low volume. Needs an explicit handling decision (exclude vs. impute) before this field is used in KPI or model calculations.
waiting_time_minutes	60	1.2%	Genuine missing values; low volume. Same handling decision applies as above.
All other variables	0	0.0%	Fully complete.

4.2 Validity and Logical Consistency

- No duplicate rows and no duplicate appointment_id values (0 found).
- previous_no_shows never exceeds previous_appointments (0 violations), consistent with the data dictionary's stated constraint.
- booking_date is never later than appointment_date (0 violations), confirming booking sequence is logically valid.
- age and age_group are fully consistent for all 5,000 records (0 mismatches).
- distance_to_clinic_km and waiting_time_minutes contain no negative values.
- Categorical fields (gender, appointment_type, appointment_time, reminder_sent/channel, appointment_outcome) all use a small, clean, well-defined set of values with no free-text entries or spelling variants.

4.3 Data Quality Issue Identified

A discrepancy was found between the appointment dataset and the Clinic Knowledge Base: the Knowledge Base states the clinic is closed on Sundays and public holidays, yet 737 appointments (14.7% of all records) are recorded with appointment_day = “Sunday.” This should be clarified with stakeholders before Sunday-related patterns are used to draw operational conclusions — it may be a synthetic data-generation artefact rather than a true reflection of clinic scheduling.

4.4 Overall Assessment

Aside from the Sunday-scheduling discrepancy and the low-volume missing values noted above, the dataset is structurally clean, internally consistent, and suitable for the descriptive and diagnostic analysis planned for this project.

5. Relevant Business Questions

- What factors are most strongly associated with a patient missing (no-showing) an appointment?
- Does sending an appointment reminder — and the channel used — make a measurable difference to attendance?
- How does the lead time between booking and the appointment date affect the likelihood of attendance?
- Do patients with a history of previous no-shows continue to miss future appointments at a higher rate?
- Does a patient's distance from the clinic influence their ability to attend?
- Are certain appointment types, days, or times associated with higher no-show or cancellation rates?

6. Potential KPIs

KPI	Definition	Value
No-Show Rate	No-Shows / Total appointments	0.4846
Attendance Rate	Attended / Total appointments	0.4628
Cancellation Rate	Cancelled / Total appointments	0.0526
Reminder Coverage Rate	Reminder sent / Total appointments	0.7268
Average Booking Lead Time	Average booking lead days	29.6386

Initial Analytical Observations

- No-Show rate is 48.5%, exceeding the attendance rate of 46.3%.
- Appointments with no reminder have a 51.4% no-show rate versus 47.4% where a reminder was sent.
- Appointments with no previous no-shows have a 43.5% no-show rate versus 55.4% among appointments with at least one previous no-show.
- Follow-up appointments have the highest no-show rate among appointment types at 51.2%.
- Appointments booked 46–60 days ahead have a 67.7% no-show rate versus 27.8% for 1–7 days.

These are descriptive associations in this dataset and should not be interpreted as proof of causation.

7. Initial Analysis Approach

The proposed approach for subsequent stages of the project is organised into four phases:

- **Phase 1 — Data Preparation:** Resolve the Sunday-scheduling discrepancy with the project team, decide how to treat the missing distance and waiting-time values, and create a clean, versioned working copy of the dataset separate from the original file.
- **Phase 2 — Exploratory & Diagnostic Analysis:** Answer each business question using descriptive statistics and cross-tabulations of appointment_outcome against the relevant variables, building on the preliminary observations in Section 6.
- **Phase 3 — KPI Calculation & Visualisation:** Formally calculate the five KPIs above, build supporting charts, and package them into a small dashboard or reporting notebook.
- **Phase 4 — Insight & Recommendation:** Summarise which factors matter most for no-shows and translate findings into practical recommendations for the clinic and inputs for the Data Science track's predictive model.

8. Assumptions, Limitations, Risks and Dependencies

8.1 Assumptions

- reminder_channel being blank is assumed to mean “no reminder was sent,” not a data collection gap, based on the consistency

8.2 Limitations

- The dataset does not indicate which of the two clinic locations (Central or Lakeside) each appointment relates to, so location-specific patterns cannot currently be analysed.
- 90 missing distance values and 60 missing waiting-time values will slightly reduce the sample available for analysis involving those two fields.
- The dataset does not include cost, insurance, or payment information, so any link between billing and no-shows cannot be explored with current resources.

8.3 Risks

- The Sunday-appointment discrepancy could lead to incorrect conclusions about weekly scheduling patterns if not clarified before analysis.
- A 48.5% no-show rate is high; if this reflects an intentionally exaggerated synthetic scenario, KPI benchmarks calculated from it should not be presented as realistic industry figures without a caveat.

8.4 Dependencies

- Further work depends on the approved appointment dataset and data dictionary; later project tracks may use the analytics findings.

8.5 Week 5 Focus

- Complete documented data preparation and cleaning.
- Calculate the selected KPIs in the analytical workflow.
- Perform deeper exploratory analysis and segmentation.
- Develop management-ready visualisations.
- Identify evidence-based insights and recommendations to support reduced missed appointments.

9. Conclusion

The supplied HealthConnect dataset provides a strong starting point for investigating appointment attendance. The initial analysis shows a substantial no-show level and identifies previous no-shows, booking lead time, reminder status, and appointment type as useful areas for deeper investigation.

The next phase should focus on validated exploratory analysis, KPI development, visual communication, and actionable recommendations while avoiding unsupported causal conclusions.

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